Chatbot to chat with medical books using Retrieval-Augmented Generation Model

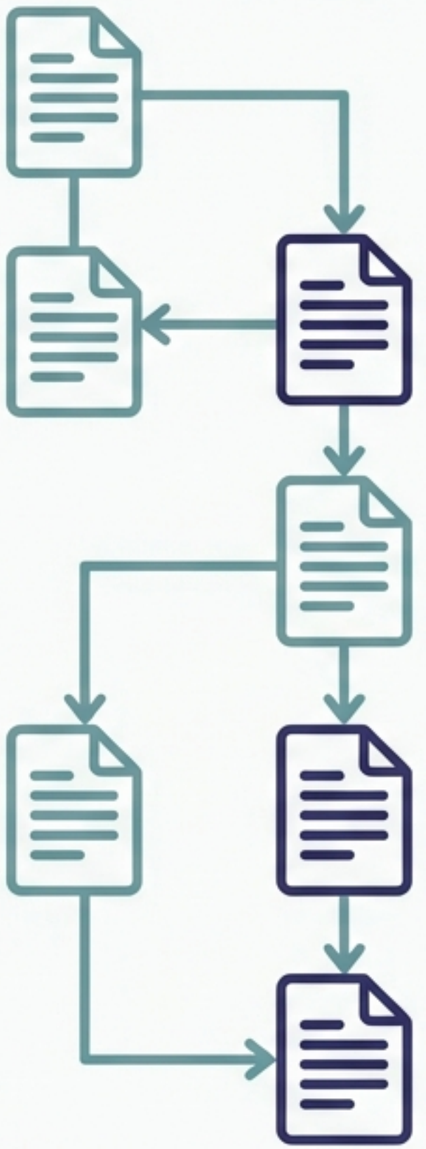


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Presentation Content

- Abstract
- Introduction
- Literature Survey
- Problem Identification
- Proposed System
- Project Objectives
- Architecture
- Algorithm
- System and Software Requirements
- Project Plan
- Future Scope & Conclusion
- References



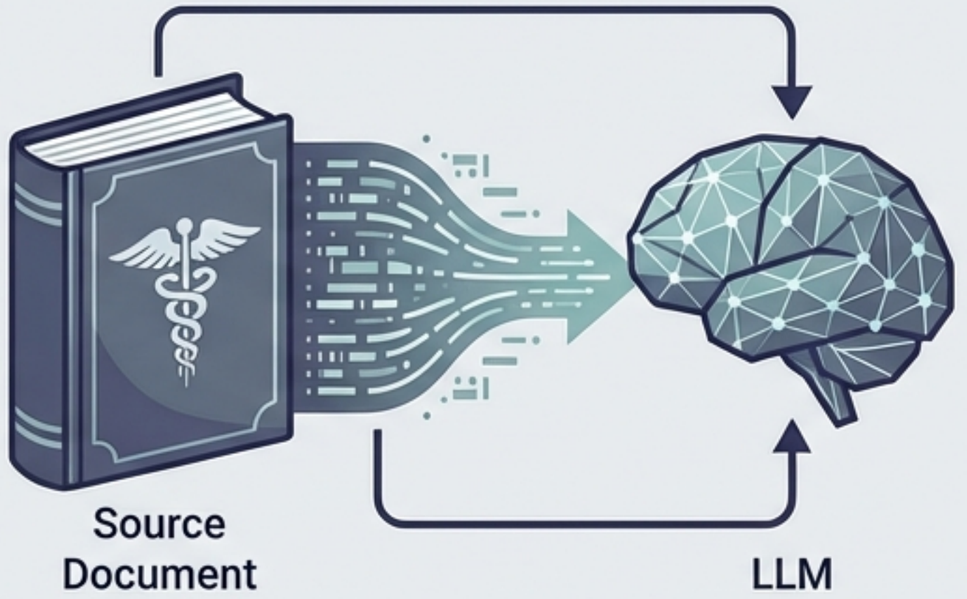
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Abstract

Core Technology: This project presents an AI-driven study assistant designed to relieve the massive reading workload of medical students. It uses a ***Large Language Model (LLM)**, which is the "brain" that understands and generates human language, powered by Google Gemini. To stop the AI from making up fake facts, we use ***RAG (Retrieval-Augmented Generation)**, which is a technique that gives the AI an "open book" to read from before it answers a question.

Technical Implementation: The system allows students to upload their heavy PDF textbooks. When a question is asked, it searches the book first, reads the exact paragraphs, and generates a grounded, highly accurate response.

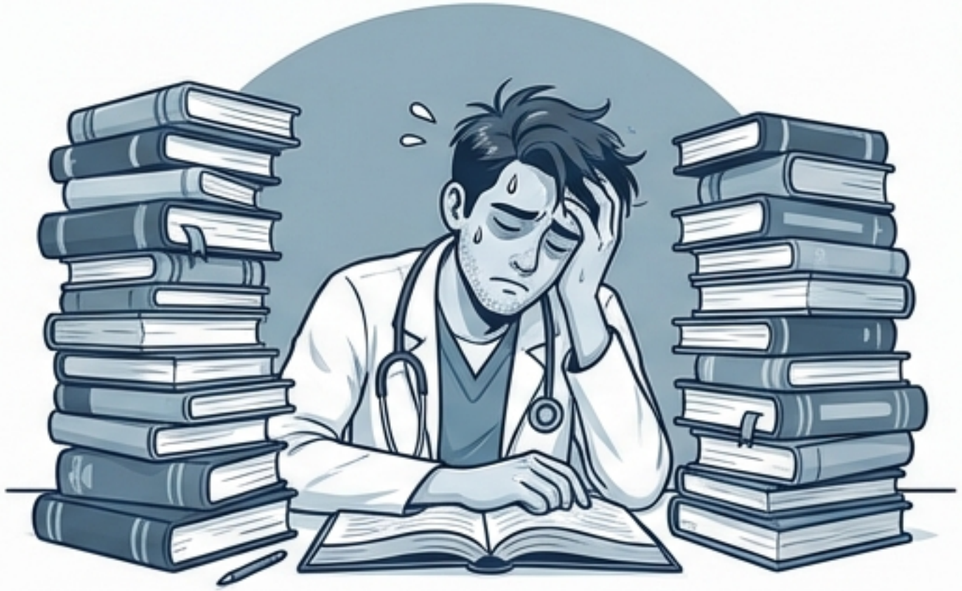
User Experience: Through a simple Streamlit web interface, students get instant, reliable medical answers backed by their own textbooks, saving hours of searching and significantly reducing study stress.



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The Need for a Reliable Medical Assistant

- Medical students face an immense workload, reading thousands of pages, which leads to high stress and fatigue.
- Standard AI tools often "hallucinate" or invent fake medical facts because they lack proper context and grounding.
- MedBuddy solves this by using the Gemini API to act as a secure, local study assistant that only answers based on uploaded files.
- **PubMed**: A free search engine accessing primarily the MEDLINE database of references and abstracts on life sciences and biomedical topics. We use this concept to source legitimate medical knowledge.
- **Meditron**: A specialized dataset concept focused specifically on high-quality, verified medical and clinical guidelines.





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Literature Survey (Recent Works)

Sl No.	Title [Year]	Author	Proposed Method	Disadvantages
1	AI-based Mock Interview System [2024]	Ayesha Sayyad	Uses AI and NLP to assess candidate responses	General AI lacks domain-specific medical grounding and factual accuracy.
2	Role of ChatGPT and AI in Medical Education [2023]	Gunther Eysenbach	Investigated standard LLMs in medical learning	High risk of authenticity issues and "hallucinations" without a retrieval system.
3	AI Chatbots in Public Health Education [2023]	Francesco Baglivo et al.	Evaluated AI chatbots for vaccination training	Lacked professional verification and reliable source-tracking for answers.

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Literature Survey (Foundational Works)

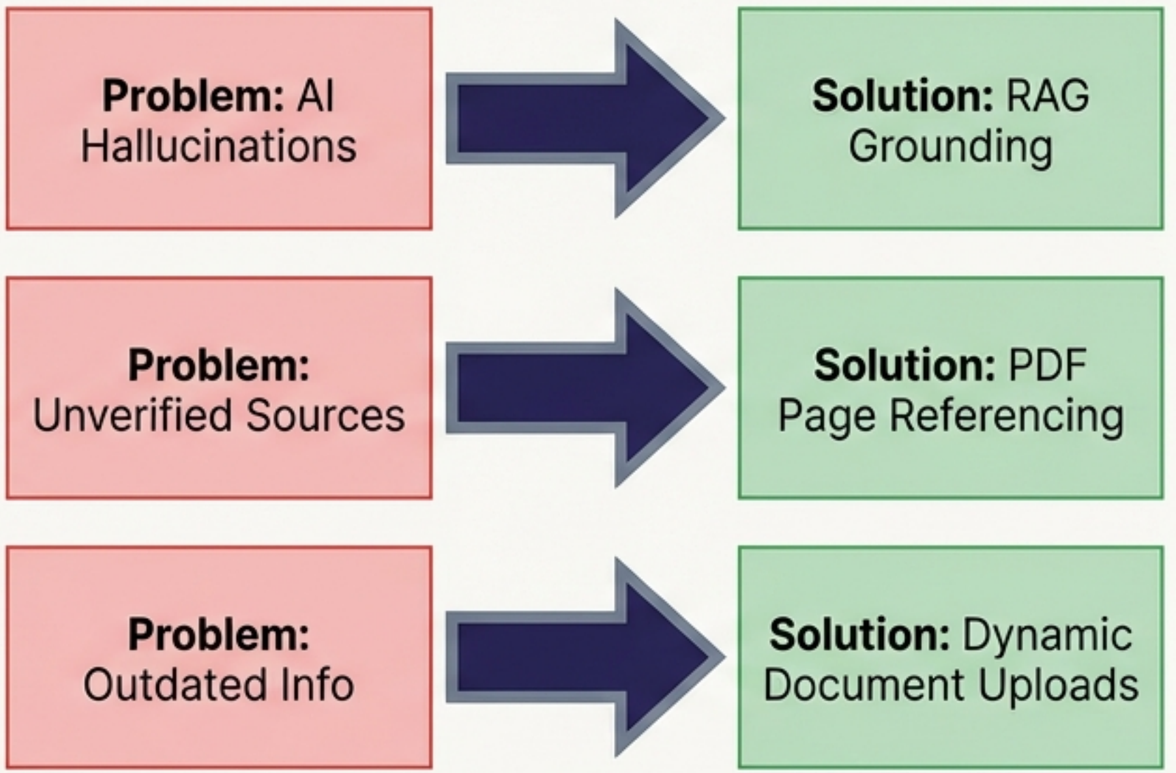
Sl No.	Title [Year]	Author	Proposed Method	Disadvantages
4	Ontology Based Personal Assistive Application [2023]	N. V. Rohil et al.	Uses ontology modeling and semantic reasoning for health	Highly complex to update with new medical textbooks and literature.
5	Farm-Haystack Question Answering System [2022]	R. S. Nambiar	QA system using pre-trained BERT models	Often lacks specificity for highly specialized medical queries.
6	ChEMBL Bot for Database [2018]	V. Murali et al.	NLP chatbot to query chemical compounds	Limited strictly to a specific chemical database, cannot read full medical PDFs.

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Problem Identification

- **Generic AI is Unsafe:** Standard AI models guess answers when they do not know them, which is highly dangerous in medical education.
- **Lack of Source Transparency:** Existing systems do not tell the student exactly which page or book the information came from.
- **Inflexible Knowledge Bases:** Hard-coded chatbots cannot easily learn from new textbook editions without complete retraining.

Problem-Solution Mapping



Problem-Solution Mapping

- **Problem:** AI Hallucinations → **Solution:** RAG
- **Problem:** AI Hallucinations → **Solution:** RAG Grounding
- **Problem:** Unverified Sources → **Solution:** PDF Page Referencing
- **Problem:** Outdated Info → **Solution:** Dynamic Document Uploads

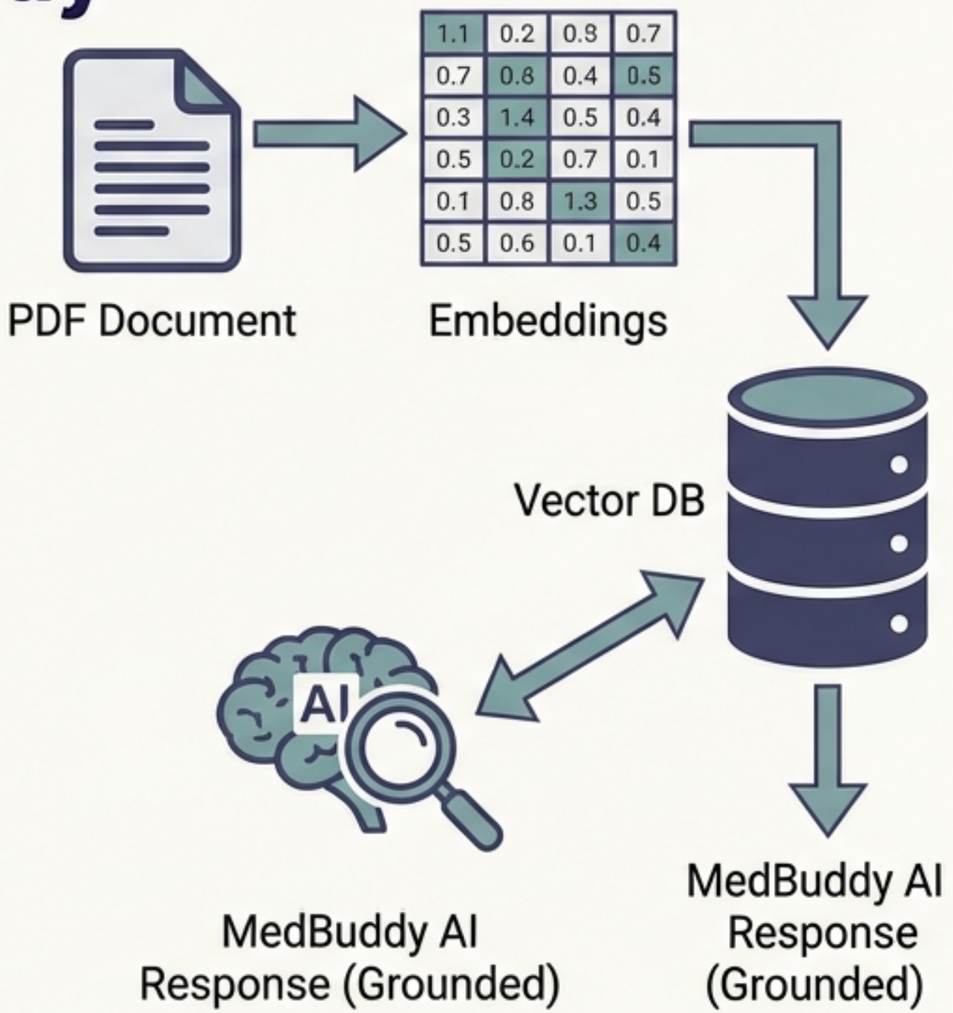
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The Proposed System: MedBuddy

MedBuddy uses Google Gemini as its core AI, enhanced by specific tools to ensure perfect medical accuracy derived from datasets like PubMed and Meditron.

Key Concepts:

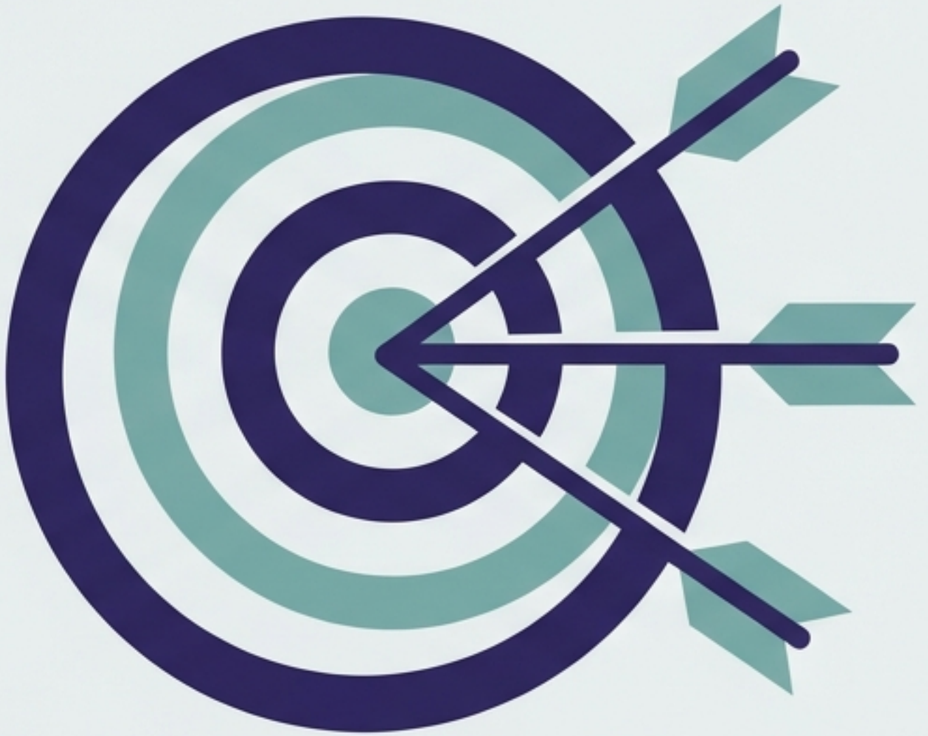
- **Embeddings**: The process of turning human words and sentences into a list of numbers so the computer can mathematically understand their meaning.
- **Vector Database**: A special storage space where we keep these "number-words." It allows the AI to instantly find paragraphs that match the user's question.
- **LLM Grounding:** A strict rule we put on the AI. We force the AI to ***only*** generate an answer using the text we provided, completely stopping it from making up outside facts.



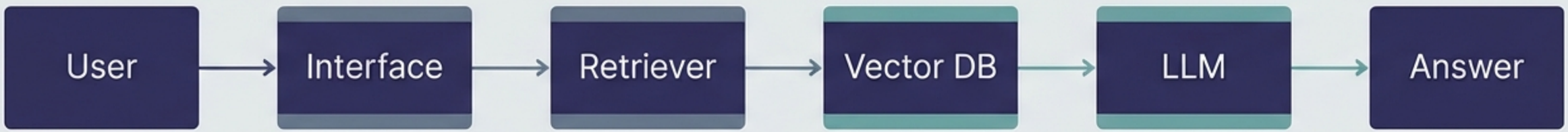
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Project Objectives

- **To build** an intuitive, user-friendly AI interface where medical students can upload massive textbook PDFs.
- **To retrieve** the exact, highly relevant textbook chunks instantly when a medical query is asked.
- **To generate** context-aware, highly accurate text responses based entirely on the retrieved data.
- **To improve** study efficiency and reduce the mental stress of medical students.
- **To evaluate** the system's accuracy and performance using the Trulens evaluation library.
- **To provide** grounded medical answers that are 100% free of AI hallucinations.



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System Architecture

The system architecture isolates the uploaded medical knowledge from the generative AI, ensuring safety and speed.

- The user interacts with the clean front-end interface.
- The system breaks the PDF into chunks and stores them mathematically.
- When asked a question, the retriever fetches only the most relevant chunks.
- The LLM reads only those chunks to formulate the final answer.

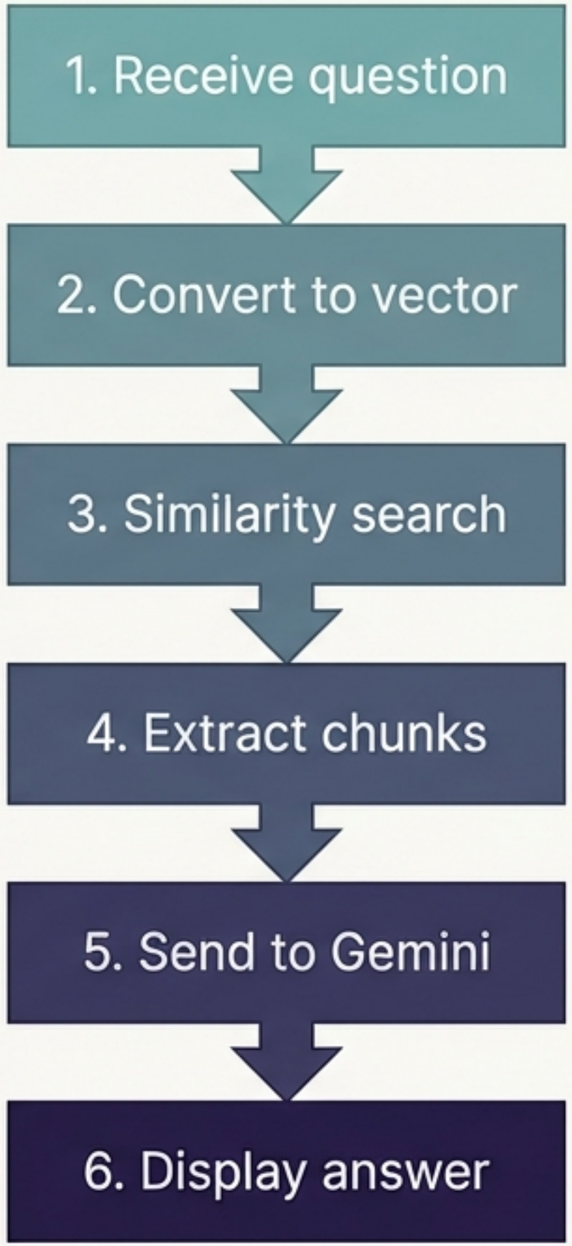
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Algorithm Logic

Query Processing: This is the act of taking the user's plain-English medical question, converting it into embeddings (numbers), and searching the Vector Database for paragraphs with the closest matching numbers.

Response Generation Flow: This is the final step where the retrieved textbook paragraphs and the original question are sent to the Gemini LLM. Gemini is instructed to merge this data into a readable, human-like answer.

1. Receive question from the user.
2. Convert question to vector embedding.
3. Perform similarity search in Vector DB.
4. Extract top matching textbook chunks.
5. Send chunks + question to Gemini LLM.
6. Display generated answer with source references.



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System and Software Requirements



Datasets & Knowledge Base

- PubMed (Medical paper
- PubMed (Medical paper formatting standard)
- Meditron (Medical data concept constraints)

Front-End

- Streamlit (Creates the web application UI)

LLM Service

- Google Gemini API

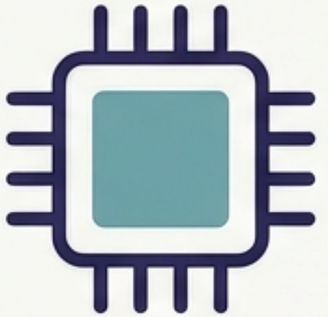
Back-End & Libraries

- Python (Core programming language)
- LlamaIndex (Framework to connect data to LLMs)
- Trulens (Library to evaluate AI groundedness)

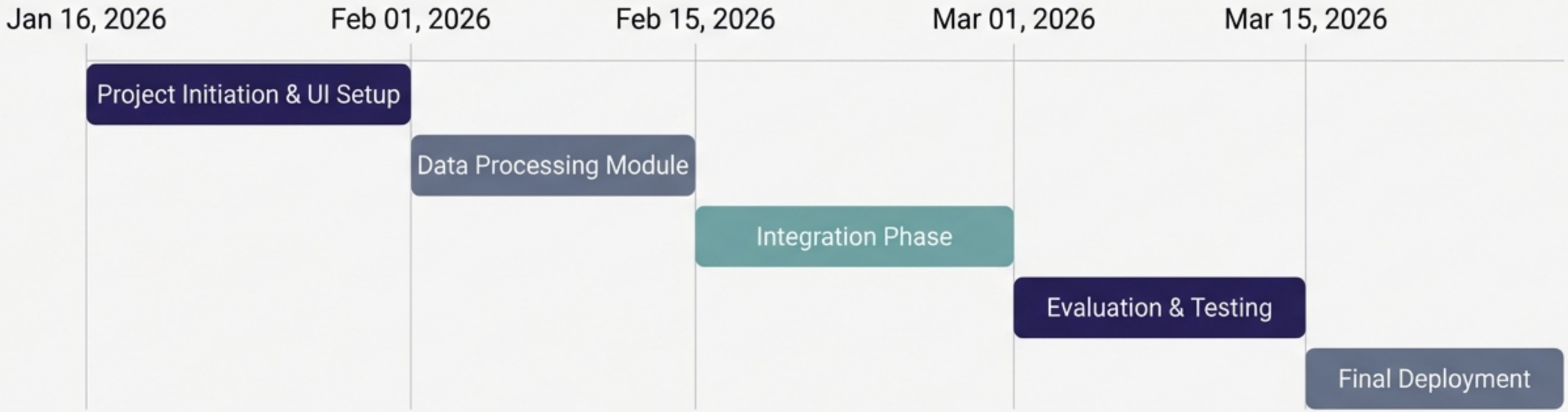


Hardware Requirements

- Processor: Intel i5 or AMD Ryzen 5 minimum
- Memory: 8GB RAM minimum
- Storage: 256GB SSD minimum



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Project Plan

- **Jan 16, 2026:** Project Initiation & UI Setup. Building the Streamlit interface and configuring Python environments.
- **Feb 01, 2026:** Data Processing Module. Implementing PDF uploading, chunking, and embedding generation.
- **Feb 15, 2026:** Integration Phase. Connecting LlamaIndex, the Vector Database, and the Gemini API.
- **Mar 01, 2026:** Evaluation & Testing. Using Trulens to test for context relevance and groundedness.
- **Mar 15, 2026:** Final Deployment. System optimization and final presentation readiness.

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Future Scope

- Expanding the system to support real-time fetching from live medical journals.
- Adding voice-to-text interaction so students can study hands-free.
- Implementing user accounts to save chat histories and create personalized study flashcards.



Conclusion

MedBuddy successfully provides a safe, grounded, and hallucination-free AI assistant. By combining Google Gemini with a rigorous RAG pipeline, it achieves high accuracy (approx. 84% groundedness). It democratizes access to complex medical knowledge, significantly reducing the mental burden on students and allowing them to study smarter, not harder.

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References

1. Gunther Eysenbach, "The Role of ChatGPT, Generative Language Models, and Artificial Intelligence in Medical Education," JMIR Publications, 2023.
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